Parental Labor Force Participation and Educational Background

An Analysis Using IPUMS CPS ASEC 2022–2025

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Research Question

What share of parents, especially mothers, with young children are out of the labor force, and does that vary by education level?

For parents of young children, it is no surprise that the decision to work is rarely straightforward. Childcare costs, salary progression, caregiving demands, early years with their kid, are just a few of the many thoughts parents might be absorbed by at that point in time. Understanding tradeoffs can be complicated too. It is not hard to imagine that these pressures also fall unevenly across different education levels and between mothers and fathers. This analysis uses nationally representative survey data to observe and examine who is out of the labor force, how education can impact that, and how the pattern has shifted between 2022 and 2025.

Data and Methods

This analysis uses the Current Population Survey (CPS) Annual Social and Economic Supplement (ASEC), accessed through IPUMS. The CPS is the primary source for labor force statistics in the United States, and the ASEC supplement adds detailed information on education, income, and household composition that makes this analysis possible.

The analysis of the population was constrained to prime working-age adults between 25 and 54. Adults under 25 are often still in school and adults over 54 begin to show early retirement patterns unrelated to childcare. Within that population, parents of young children are identified as anyone with at least one own child under age 5 in the household.

ASECWT, the person-level survey weight provided by IPUMS for ASEC records, is used to weight all estimates. This ensures estimates are representative of the U.S. population, rather than just the survey sample.

Key Variables

Labor force status is derived from EMPSTAT, which captures whether a person is working, unemployed, or not in the labor force (NILF). The NILF category for this analysis refers to people who are neither working nor actively looking for work.

Educational background is derived from EDUC and boxed into four categories: less than high school, high school diploma, some college, and bachelor's degree or higher.

Parental status is derived using NCHLT5, which counts the number of own children under age 5 in the household. Anyone with at least one child under 5 is classified as a parent of a young child for this analysis.

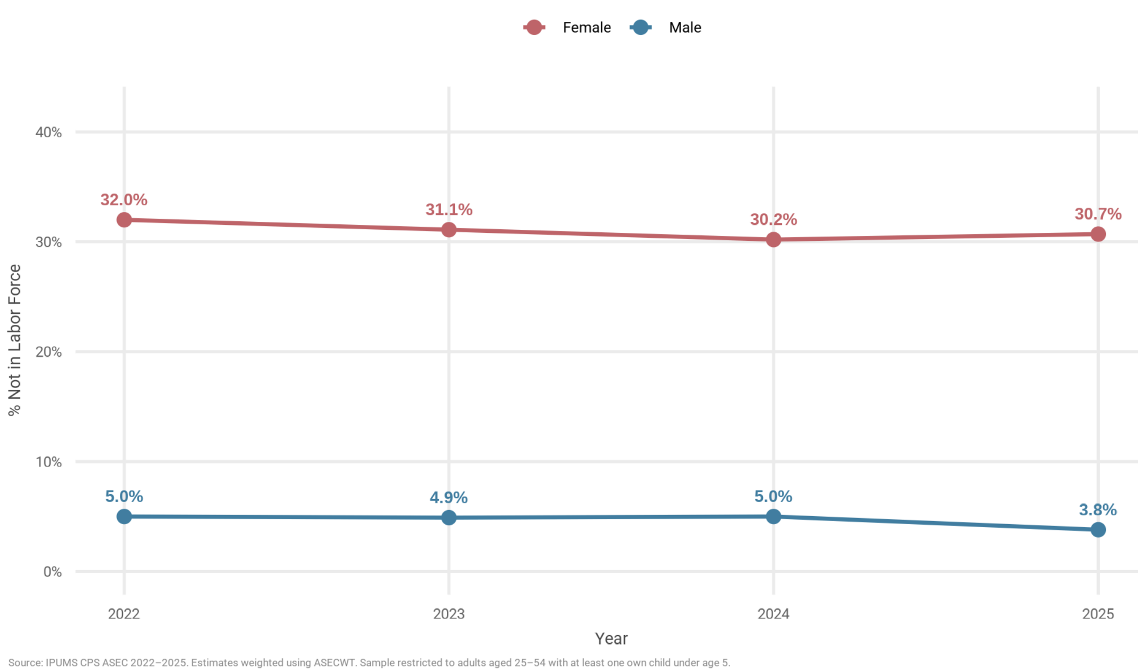
Finding 1: The Gender Gap Persists

In [43]:

```
from IPython.display import Image

Image("outputs/figures/nilf_trend_by_sex.png")
```

Out[43]:

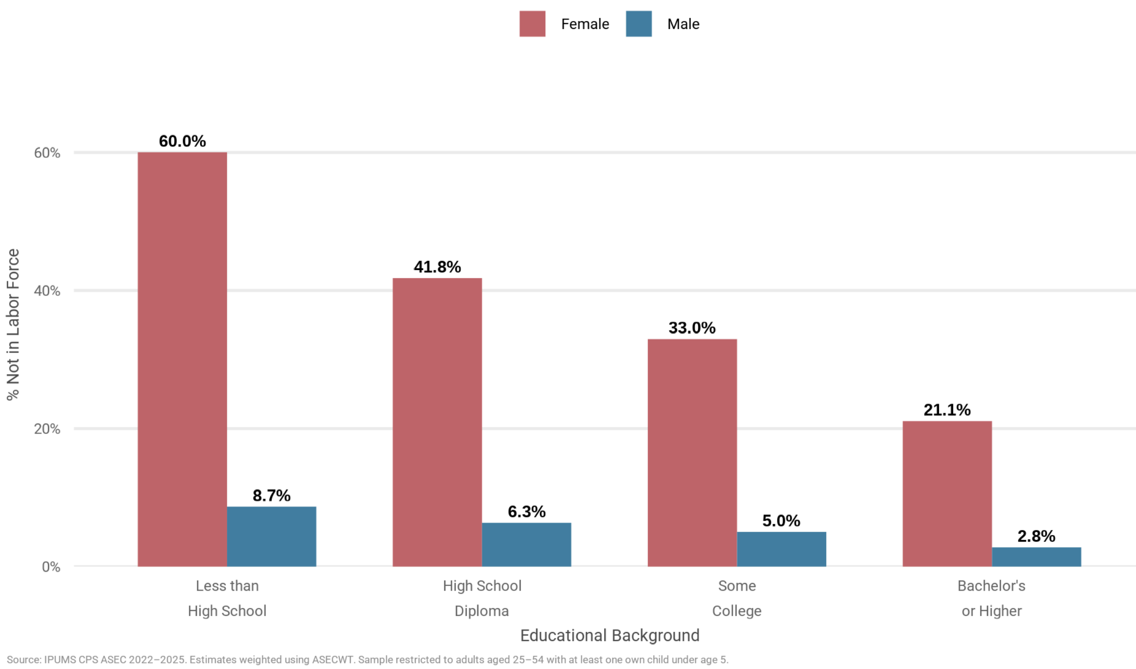


This was the first visual I built. And it was clear that between 2022 and 2025, mothers of young children have consistently been around 30% not in the labor force. It surprised me that this was roughly six times the rate of fathers, who hover between 4-5%. The gap seems to have barely moved over four years, suggesting this isn't a temporary post-COVID pattern, but a persistent structural one. I decided to look deeper to see if education revealed more to this story.

Finding 2: Education Correlates with the Gap

In [44]:
```python
Image("outputs/figures/nilf_by_educ_sex.png")
```

Out[44]:

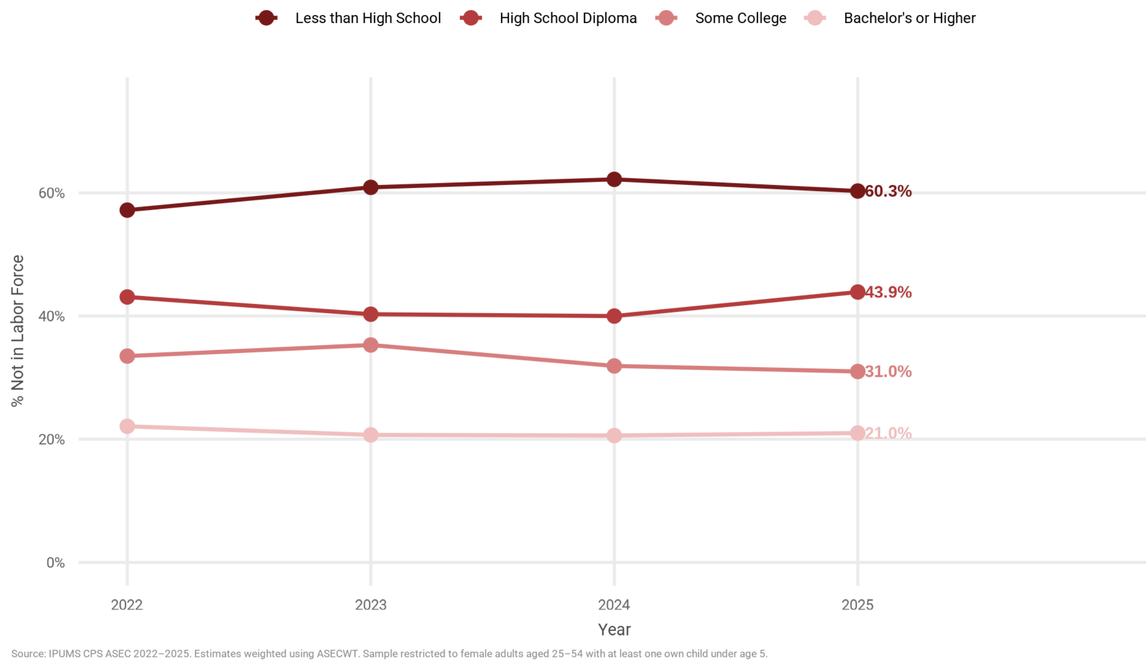


Right away, it was apparent that labor force participation for mothers varies significantly by education, but in comparison, significantly less so for fathers. Among mothers with less than a high school diploma, 60% are out of the labor force. That drops to 21% among mothers with a bachelor's degree or higher. This is still a significant share, but paints a different picture. For fathers, the range runs from 8.7% to 2.8% across the same education groups. The gap between mothers and fathers is widest at the lowest education levels and narrows as education level increases.

Finding 3: Stable differences in recent years

In [45]:
```python
Image("outputs/figures/nilf_mothers_by_educ.png")
```

Out[45]:

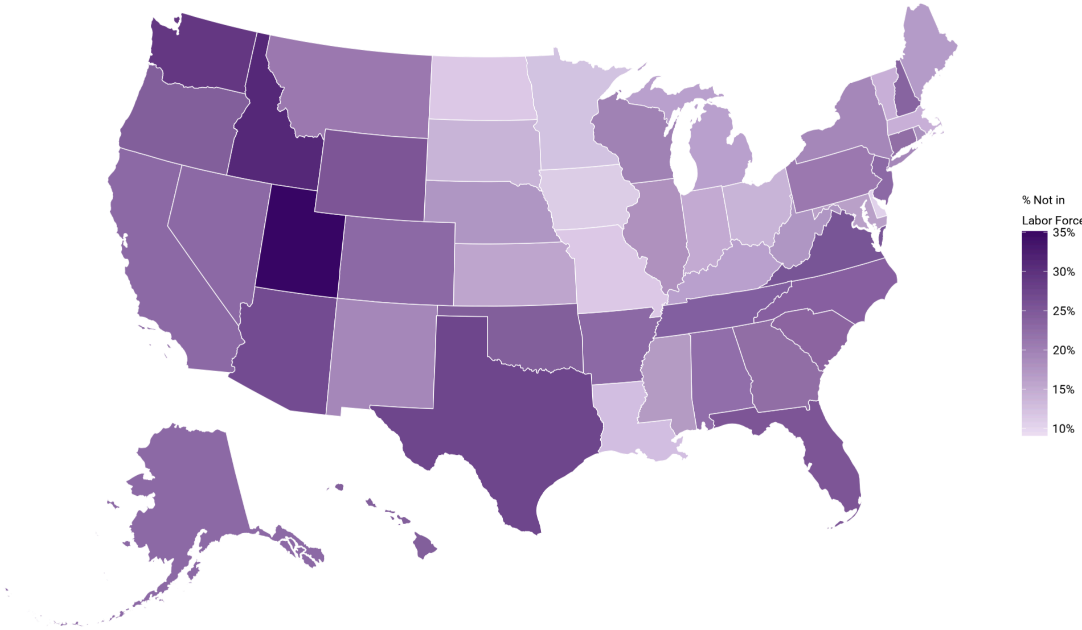


Looking at mothers alone over time, we see level differences by education. Mothers without a high school diploma maintain a high NILF rate, hovering around 60% across all four years. Mothers with a bachelor's degree or higher sit around 21%, with little change. The distance between the most and least educated mothers has remained stable, suggesting that the underlying causes for education-level differences in labor force participation have not changed much in recent years.

Finding 4: Geography Matters Even Among the Most Educated

In [46]:
```
Image("outputs/figures/nilf_map_bachelors_mothers.png")
```

Out[46]:



Source: IPUMS CPS ASEC 2022–2025. Estimates weighted using ASECWT. Sample restricted to female adults aged 25–54 with at least one own child under age 5.

Even among the most educated mothers, those with a bachelor's degree or higher, participation looks very different across the nation. Utah stands out at 35%, the highest NILF rate in the country for this group. DC sits at the opposite end at 9%. That's a 26 percentage point gap among women with the same level of education. Several Midwestern states have among the lowest NILF rates for college-educated mothers, a geographic pattern I was previously unaware of. The spatial variation also makes me curious about how differences in childcare infrastructure and cultural norms play varying depending on where you live.

Limitations and Next Steps

- **Causality**: The association between education and labor force participation does not tell us whether education causes higher participation, or whether other factors correlated with education are driving the pattern.
- **Geography**: ASEC data supports state-level estimates but not county or metro-level analysis. The geographic map masks meaningful variation within states.
- **Reason for NILF**: At present, we cannot differentiate between someone who left voluntarily to care for a child and someone who wanted to work, but couldn't find affordable childcare. Thinking about individual decisions would be important for future work.
- **Sample size**: Some states have small samples for the bachelor's degree and higher mothers subgroup, which makes those estimates less reliable. For example, while

most states had greater than 100 observations, Rhode Island (n=94), Maine(n=80), and Kentucky (n=79) seem to be worth calling out. The estimates should be regarded with caution.

- **What I'd be curious to dive deeper into**: I'd be interested in adding childcare cost data at the state level to test whether states with higher childcare costs also have higher maternal NILF rates. This could also help us determine whether that relationship is stronger at lower education levels where wages are lower relative to childcare costs. I'm also curious about what happens after the early years: do mothers who left the labor force rejoin once their children reach school age, and does that recovery look different depending on education level or state? CPS has variables that would let you track NILF rates by age of youngest child, which could start to answer that question. However, a longer term study, perhaps with a natural experiment could help with evaluating specific state policy. Topics around paid family leave or subsidized childcare expansion would be interesting to explore further.

Replication

Data: IPUMS CPS

To replicate, please register at IPUMS and extract the following variables for ASEC samples 2022–2025: `AGE, SEX, RACE, HISPAN, EDUC, EMPSTAT, INCWAGE, WTFINL, ASECWT, NCHLT5, RELATE, STATEFIP`

All code available at: github.com/nehakmvlli/parental-lfp-education-cps

Data Citation

Sarah Flood, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, Daniel Backman, Etienne Breton, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, David Van Riper, and Kari C.W. Williams. IPUMS CPS: Version 13.0 [dataset]. Minneapolis, MN: IPUMS, 2025. https://doi.org/10.18128/D030.V13.0

In [ ]:

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